

Day 64 - Runtime Health Dashboard

Date: 2026-05-11

Lane: Runtime / deployment / security

Status: developer-side PASS, production owner visual status unchanged

Blocker class: NONE for local runtime-health proof

Objective

Expose a safe runtime health contract in Mission Control so the owner-facing dashboard can show the current runtime commit, local bind posture, standalone bundle status, PID/listener match, and deployment timestamps without exposing secrets or raw local paths.

Scope

Implemented a protected runtime health endpoint and wired its safe payload into the Mission Control dashboard data model.

The endpoint is intentionally read-only. It does not restart services, does not mutate environment files, and does not execute external writes.

Files Changed

- 'src/lib/runtime-health.ts'
- 'src/app/api/runtime/health/route.ts'
- 'src/app/api/runtime/health/route.test.ts'
- 'src/app/api/status/route.ts'
- 'src/components/dashboard/dashboard.tsx'
- 'src/components/dashboard/widget-primitives.tsx'
- 'src/components/dashboard/widgets/runtime-health-widget.tsx'
- 'runtime/day-64-runtime-health-live.json'
- 'runtime/day-64-dashboard-runtime-health-live.json'
- 'runtime/day-64-runtime-health-dashboard.md'
- 'runtime/day-64-runtime-health-dashboard.pdf'

Routes Changed

- Added protected 'GET /api/runtime/health'.
- Extended 'GET /api/status?action=dashboard' with 'runtimeHealth'.

UI Behavior

Mission Control dashboard now has a runtime health widget that shows:

- Runtime status.
- Current source commit and branch.
- Expected safe bind host and port.
- Standalone bundle/static readiness.
- PID file and listener match state.
- Last build/static sync timestamps.
- Exact blocker list when the runtime cannot be classified as live.

No raw local filesystem paths are shown. The PID file is reported as a safe relative label only.

Service / Runtime Behavior

Runtime health detection checks:

- '.next/standalone/server.js' or standalone 'server.js'.
- '.next/standalone/.next/static' or '.next/static'.
- '.next/standalone/server.pid' or 'server.pid'.
- Local listener PID on the configured port.

- Whether the PID file matches the active listener.

The bind contract remains local-only:

- Expected host: '127.0.0.1'
- Override variable: 'MC_HOSTNAME'
- Proof port: '3337'
- No new public exposure added: 'true'

Proof Artifacts

- 'runtime/day-64-runtime-health-live.json'
- 'runtime/day-64-dashboard-runtime-health-live.json'

Observed live proof:

- Runtime health status: 'LIVE'
- Runtime source commit at proof time: 'ccc0ce5'
- Branch: 'to-knowledge-mc'
- PID file value: '81685'
- Listener PID: '81685'
- PID matched listener: 'true'
- Runtime cwd mode: 'standalone'
- Raw path leak in dashboard proof artifact: 'false'
- Secret output in dashboard proof artifact: 'false'

Tests / Checks

Fresh validation required before commit:

- 'git diff --check'
- 'pnpm test src/app/api/runtime/health/route.test.ts'
- 'pnpm run typecheck'
- 'pnpm run build'
- 'pnpm test'
- 'node scripts/check-protected-file-invariants.mjs'
- staged secret scan
- '.env' diff check

Deploy / Restart / Smoke

Mission Control standalone was rebuilt and restarted in local-only mode on '127.0.0.1:3337'.

Live smoke results:

- 'GET /api/runtime/health' authenticated with API key and session cookie returned 'status: LIVE'
- 'GET /api/status?action=dashboard' authenticated included 'runtimeHealth'.
- Runtime health proof confirmed no public bind.

Rollback

Use:

```
""bash
git revert <day-64-runtime-health-commit-sha>
MC_HOSTNAME=127.0.0.1 PORT=3337 bash scripts/start-standalone.sh
""
```

Safety Confirmation

- No '.env' changes.
- No secrets printed.
- No auth weakening.

- No public local exposure added.
- No external writes.
- No fake runtime status.
- No raw local paths in staged runtime proof artifacts.

Next Day

Day 65 starts automatically after commit and push: Protected Route Smoke 100% Closure.